PHASE 6 – PROJECT TESTING

AI FAQ Assistant API – AI-Powered FAQ and Customer Support Automation

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1. Introduction

Testing is an important phase of the **AI FAQ Assistant API** project. The purpose of testing is to verify that the developed REST API functions correctly, handles valid and invalid requests, protects secured endpoints, and provides appropriate responses.

The project was tested using two approaches:

- **Automated API integration testing** using the test_api.js file.
- **Manual API testing** using the provided Postman collection.

The testing covers user authentication, FAQ management, FAQ searching, AI services, validation, authorization, and error handling.

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2. Testing Objectives

The main objectives of testing are:

1. To verify that the API server starts correctly.
2. To verify user registration and login.
3. To verify JWT-based authentication.
4. To verify FAQ creation, retrieval, updating, and deletion.
5. To verify FAQ search functionality.
6. To verify protected API endpoints.
7. To verify AI answer generation.
8. To verify AI-generated FAQ functionality.
9. To verify invalid request and error handling.
10. To ensure that test data can be cleaned after automated testing.

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3. Testing Environment

The following environment was used for testing the application.

Component	Testing Environment
Application Type	RESTful Backend API
Runtime	Node.js
Framework	Express.js
Database	MongoDB
ODM	Mongoose
Authentication	JWT
Password Security	bcrypt
AI Service	Google Gemini API
API Testing Tool	Postman
Automated Testing	test_api.js
API Base URL	http://localhost:5000
Data Format	JSON

The API can be tested locally after configuring the required environment variables such as MongoDB connection details, JWT secret, and Gemini API key.

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4. Testing Methodology

The project follows a functional API testing approach.

The testing process consists of the following steps:

Step 1: Start MongoDB and the Node.js application.

Step 2: Verify the API health endpoint.

Step 3: Test user registration.

Step 4: Test duplicate email prevention.

Step 5: Test user login and obtain the JWT token.

Step 6: Use the JWT token to access protected endpoints.

Step 7: Test FAQ CRUD operations.

Step 8: Test FAQ search.

Step 9: Test AI answer and AI FAQ generation services.

Step 10: Test invalid input and error handling.

Step 11: Verify that temporary automated test records are removed after testing.

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5. Authentication Testing

Authentication testing verifies the user registration, login, JWT token generation, and protected profile functionality.

Test ID	Test Case	Request	Expected Result
AUTH-01	User Registration	POST /api/auth/register	User should be registered successfully
AUTH-02	Duplicate Registration	Register using existing email	Request should be rejected
AUTH-03	User Login	POST /api/auth/login	JWT token should be returned
AUTH-04	Profile Access	GET /api/auth/profile with JWT	User profile should be returned
AUTH-05	Protected Access	Profile without valid JWT	Access should be rejected

The automated testing file specifically checks registration, duplicate registration prevention, login, and access to the protected profile endpoint.

The Postman collection also contains separate requests for:

- User Registration
- User Login
- User Profile

The login request automatically stores the returned JWT token in the Postman environment for use in subsequent protected requests.

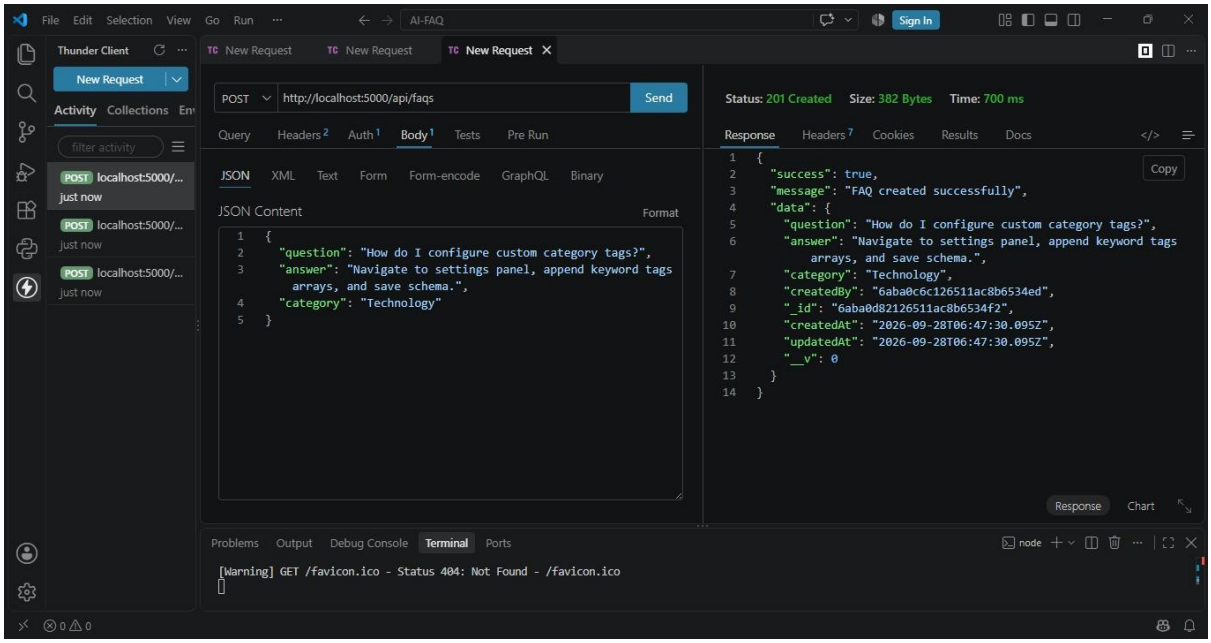


Figure – User Login and JWT Authentication Test

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6. FAQ Management Testing

The FAQ module provides CRUD operations for managing frequently asked questions.

The following operations were tested:

Test ID	Test Case	Method	Endpoint
FAQ-01	Create FAQ	POST	/api/faqs
FAQ-02	Get All FAQs	GET	/api/faqs
FAQ-03	Get FAQ by ID	GET	/api/faqs/:id
FAQ-04	Update FAQ	PUT	/api/faqs/:id
FAQ-05	Delete FAQ	DELETE	/api/faqs/:id
FAQ-06	Invalid FAQ ID	GET	/api/faqs/invalid-id-format
FAQ-07	Search FAQ	GET	/api/faqs/search?q=Express

The create, update, and delete operations require authentication. FAQ retrieval and search are publicly accessible.

The FAQ model supports the following categories:

- Technology
- Education
- Health
- Banking
- General

The automated test file creates a temporary FAQ, retrieves it, searches for it, updates it, and finally deletes it.

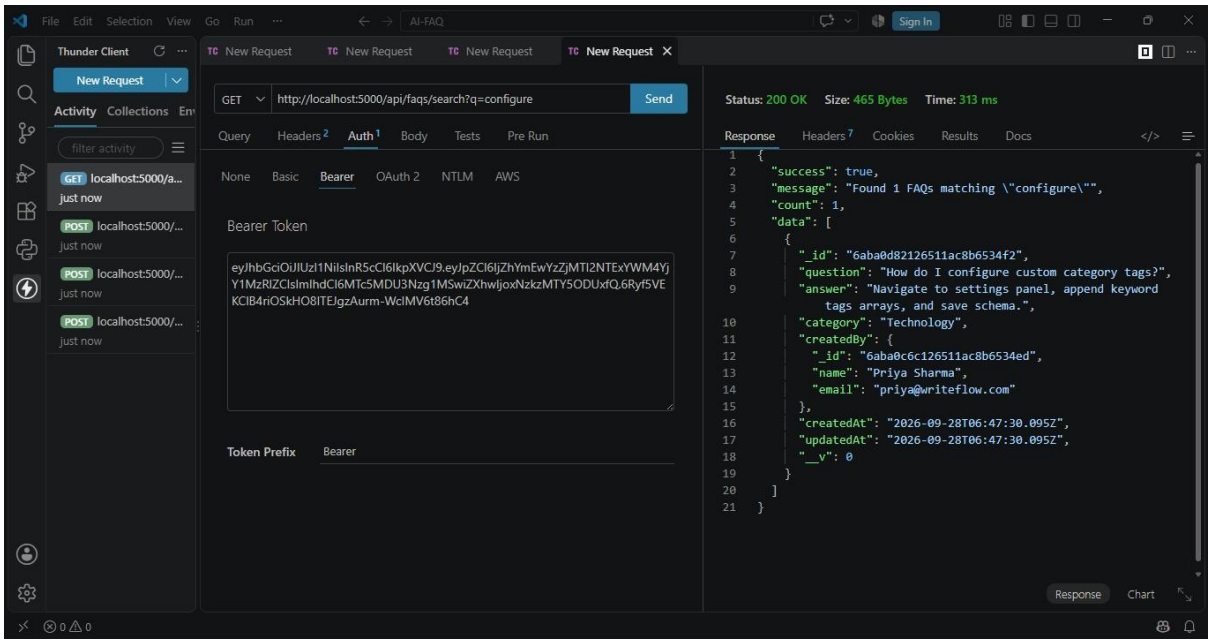


Figure – FAQ Search API Test Result

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7. Authorization and Security Testing

Security testing verifies that protected resources cannot be accessed without proper authentication.

JWT authentication is used to protect selected API endpoints.

The testing verifies:

- Valid JWT token allows access.
- Protected requests contain the Authorization: Bearer <token> header.
- FAQ modification operations require authentication.
- FAQ update and deletion are restricted to the creator of the FAQ.
- Invalid requests are processed through the application's error-handling mechanism.

The Postman collection uses a reusable token environment variable for authenticated requests.

The automated test also obtains a JWT token during login and uses it for subsequent protected API requests.

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8. AI Service Testing

The project provides two AI-based services through Google Gemini integration.

8.1 AI Answer Generator

Endpoint:

POST /api/ai/answer

Example request:

```
{
  "question": "What is Artificial Intelligence?"
}
```

The endpoint receives a question and sends it to the configured Gemini AI service to generate an answer.

8.2 AI FAQ Generator

Endpoint:

POST /api/ai/generate-faq

Example request:

```
{
  "topic": "MongoDB"
}
```

The AI service generates an FAQ response based on the supplied topic.

The automated test checks whether the AI service responds successfully when the API key is configured. It also verifies that AI-related errors are handled gracefully when the service cannot successfully complete the request.

Test ID	Test Case	Expected Result
AI-01	AI Answer Request	AI-generated answer or formatted error
AI-02	AI FAQ Generation	Structured FAQ response or formatted error
AI-03	AI Service Error	Error should be handled gracefully

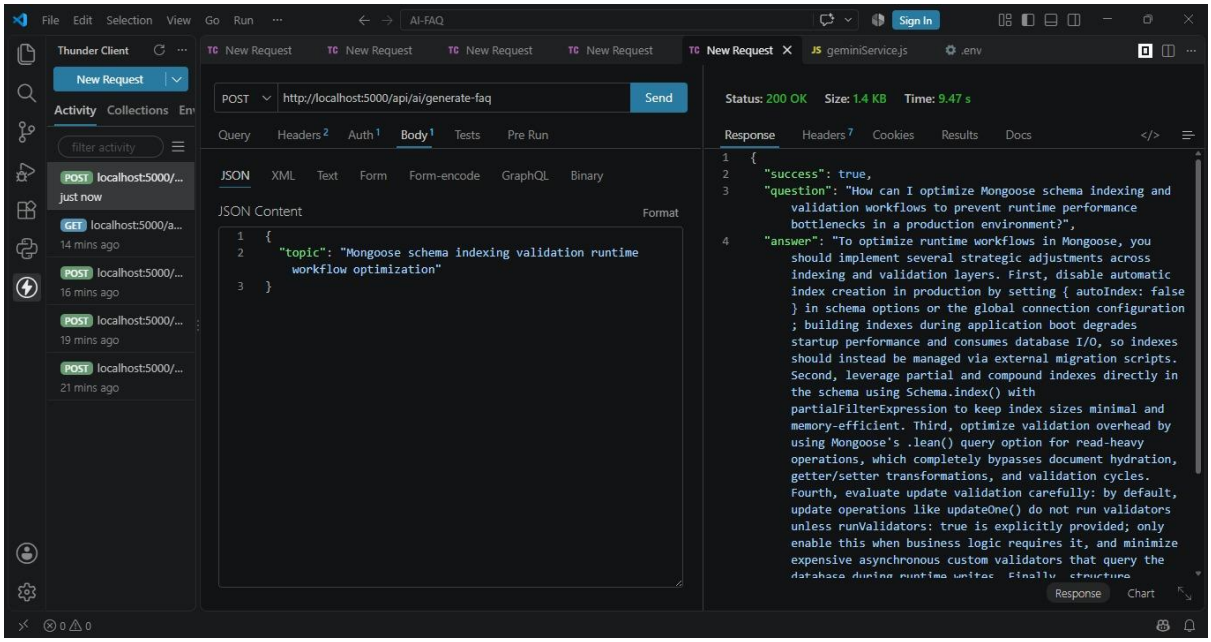


Figure – AI FAQ Generation Test Result

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9. Validation and Error Handling Testing

The application includes validation middleware and centralized error handling.

Testing was performed for situations such as:

- Duplicate user registration.
- Invalid FAQ ID.
- Invalid or unsuccessful API requests.
- Authentication-related failures.
- Database validation errors.
- AI service/API key errors.

The automated testing script specifically verifies duplicate registration and invalid FAQ ID handling.

The application also contains centralized error handling for Mongoose validation errors, invalid MongoDB object IDs, and duplicate-key errors.

Test ID	Error Scenario	Expected Behaviour
ERR-01	Duplicate email	Request rejected with error response
ERR-02	Invalid FAQ ID	Appropriate error response
ERR-03	Invalid authentication	Protected request rejected
ERR-04	Database validation error	Formatted error response
ERR-05	AI service failure	Error handled without application crash

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10. Automated API Integration Testing

The project contains a dedicated file named:

test_api.js

This file performs integration testing against the application.

The automated testing sequence includes:

1. MongoDB connection.
2. API health check.
3. User registration.
4. Duplicate registration check.
5. User login.
6. Protected profile request.
7. FAQ creation.
8. FAQ retrieval.
9. FAQ retrieval using ID.
10. Invalid FAQ ID test.
11. FAQ search.
12. FAQ update.
13. AI answer service check.
14. AI FAQ generation check.

15. FAQ deletion.
16. Test-data cleanup.

The script creates temporary test records using a dynamically generated email address and removes the test user and FAQ after the test sequence.

This helps prevent unnecessary test data from remaining in the database.

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11. Postman Testing

A Postman collection named **AI FAQ Assistant API** is included with the project.

The collection is divided into three major sections:

Auth

- User Registration
- User Login
- User Profile

FAQs

- Create FAQ
- Get All FAQs
- Get FAQ by ID
- Update FAQ
- Delete FAQ
- Search FAQs

AI Services

- AI Answer Generator
- AI FAQ Generator

The collection uses:

{{baseUrl}}

as the base URL and:

{{token}}

for JWT authentication.

The login request contains a Postman test script that automatically saves the JWT token after successful login.

The Create FAQ request similarly stores the generated FAQ ID for later requests.

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12. Testing Summary

The testing phase covers the major functional components of the AI FAQ Assistant API.

Module	**Testing Coverage**
API Health Check	Tested
User Registration	Tested
Duplicate Registration	Tested
User Login	Tested
JWT Authentication	Tested
User Profile	Tested
FAQ Creation	Tested
FAQ Retrieval	Tested
FAQ Update	Tested
FAQ Deletion	Tested
FAQ Search	Tested
Invalid FAQ ID Handling	Tested
AI Answer Service	Tested
AI FAQ Generation	Tested
Error Handling	Tested
Automated Integration Testing	Implemented
Postman API Testing	Implemented

The automated test script provides a repeatable sequence for checking the major API functions, while the Postman collection provides a convenient interface for manually testing individual endpoints.

13. Testing Conclusion

The testing phase validates the functional components of the **AI FAQ Assistant API**. Authentication, FAQ management, FAQ search, protected endpoints, AI services, validation, and error-handling scenarios are covered through automated integration testing and Postman-based API testing.

The inclusion of both test_api.js and the Postman collection provides two complementary methods for verifying the API during development and future maintenance.